

Basic Principles of Medicine 1
Module: Foundation to Medicine | Lecture 24

Apoptosis and Necrosis

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Objectives

Define apoptosis	SOM.MK.I.BPM1.1.FTM.3.HCB.o819
Define necrosis	SOM.MK.I.BPM1.1.FTM.3.HCB.o820
State the importance of apoptosis	SOM.MK.I.BPM1.1.FTM.3.HCB.o821
State the importance of necrosis	SOM.MK.I.BPM1.1.FTM.3.HCB.o822
Discuss some common causes of cellular injury.	SOM.MK.I.BPM1.1.FTM.3.HCB.o823
Compare reversible vs. irreversible injuries.	SOM.MK.I.BPM1.1.FTM.3.HCB.o824
Describe the early cellular responses to injury.	SOM.MK.I.BPM1.1.FTM.3.HCB.o825
Discuss some examples of physiological processes that lead to cell death by apoptosis	SOM.MK.I.BPM1.1.FTM.3.HCB.o826
Discuss some examples of pathological processes that lead to cell death by necrosis	SOM.MK.I.BPM1.1.FTM.3.HCB.o827
Discuss the morphological changes associated with cell death by apoptosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o828
Discuss the morphological changes associated with cell death by necrosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o829
Discuss the biochemical features associated with cell death by apoptosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o830
Discuss the biochemical features associated with cell death by necrosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o831
Describe the intrinsic apoptosis pathway.	SOM.MK.I.BPM1.1.FTM.3.HCB.o832
Describe the extrinsic apoptosis pathway.	SOM.MK.I.BPM1.1.FTM.3.HCB.o833
Describe the role of caspases in apoptosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o834
Discuss methods to assess apoptosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o835
Discuss methods to assess necrosis.	SOM.MK.I.BPM1.1.FTM.3.HCB.o836

Recommended Reading

- Lippincott's Illustrated Reviews: Cell & Molecular Biology, 2nd edition
- Chandler & Viselli 2019
- Chapter 22: Abnormal Cell Growth Pg. 221-230
- <https://meded.lwwhealthlibrary.com/content.aspx?sectionid=181789589&bookid=2335>
- Chapter 23: Cell Death Pg. 231-243
- <https://meded.lwwhealthlibrary.com/content.aspx?sectionid=181789669&bookid=2335>

Robbins & Cotran Pathologic Basis of Disease (10th ed.)
Vinay, K., Abul, A., & Jon, A. 2021.

Chapter 2: Cell Injury, Cell Death, and Adaptations pp 33-61. <https://clinicalkeymeded.elsevier.com/books/9780323531139>

Chapter 7: Neoplasia. pp 267-276, 284-5. <https://clinicalkeymeded.elsevier.com/books/9780323531139>

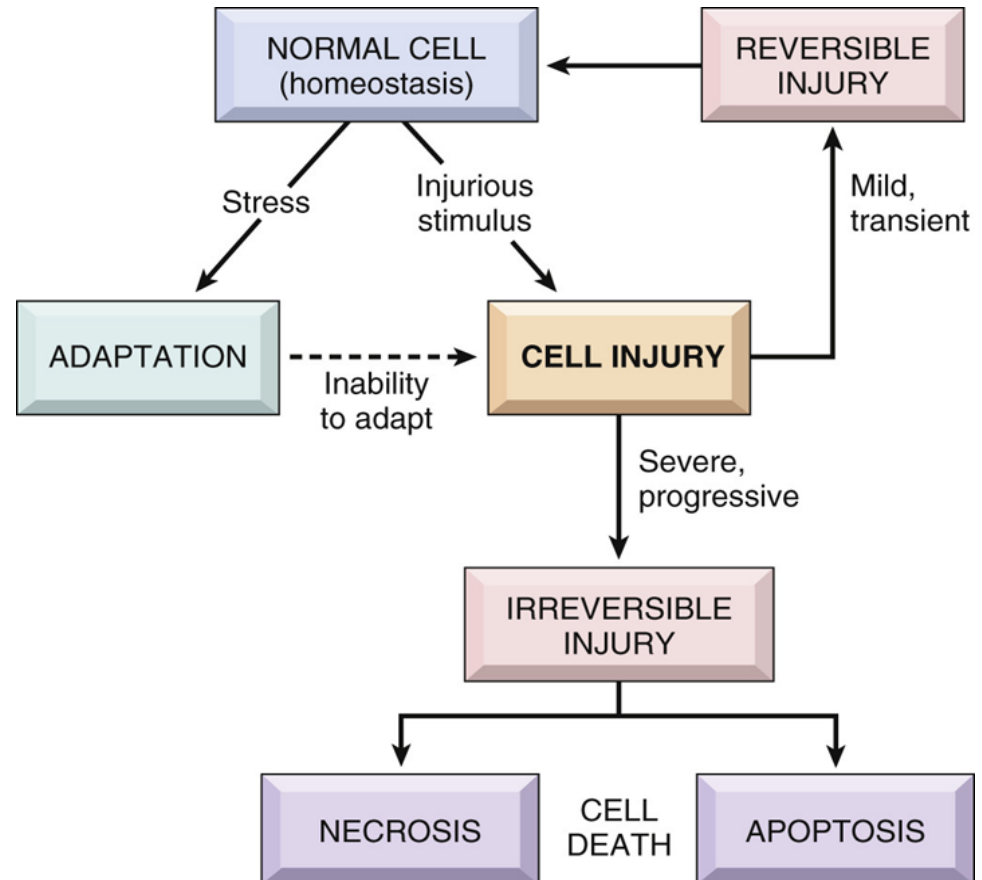
Stages of Cellular Response To Stress And Injurious Stimuli

Normal cell

-Narrow range of function and structure controlled by its state of metabolism, differentiation, specialization, availability of metabolic substrates, constraints of neighboring cells

-Physiologic demands = healthy steady state (homeostasis)

-Adapt in response to changes in the environment



Common Causes of Cellular Injury

- Oxygen deprivation
- Physical agents
- Chemical agents and drugs
- Infectious agents
- Immunological reactions
- Genetic abnormalities
- Nutritional imbalances



Progression Of Cell Injury

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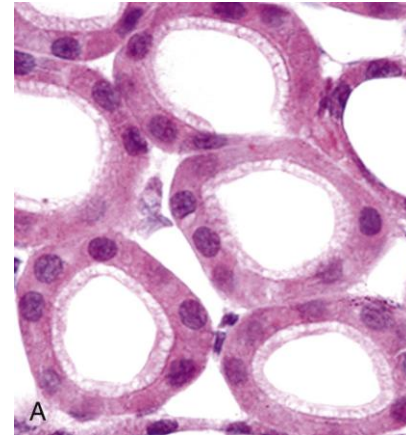
SOM.MK.I.BPM1.1.FTM.3.HCB.o825

Reversible cell injury

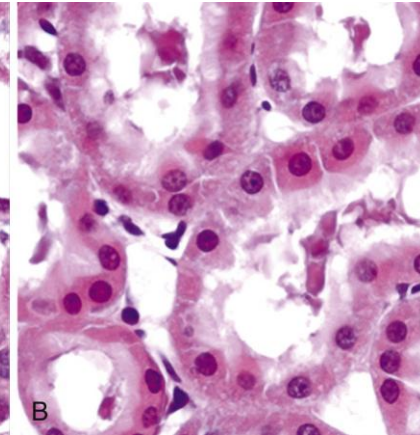
- Swelling - cells and organelles (mitochondria, ER)

Ultrastructural changes of reversible injury

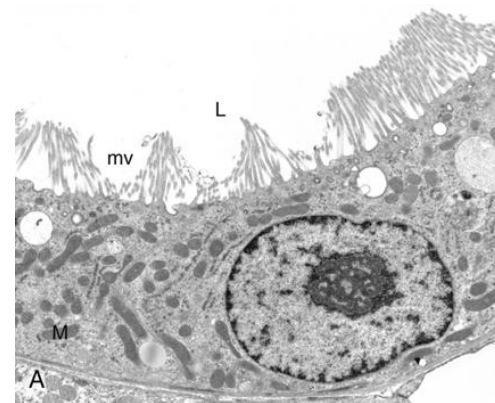
- Plasma membrane - blebs, blunting or loss of microvilli, "Myelin figures" in the cytoplasm
- ER- dilation and detached ribosomes
- Mitochondria- swelling, amorphous densities
- Nucleus- disaggregation of nuclear granular and fibrillar elements
- Cytoplasm-increased eosinophilia, "vacuolar degeneration/hydronic change"
- Fatty change-disruption of metabolic pathways and TG accumulation with lipid vacuoles in the cytoplasm e.g., hepatocytes and myocardial cells



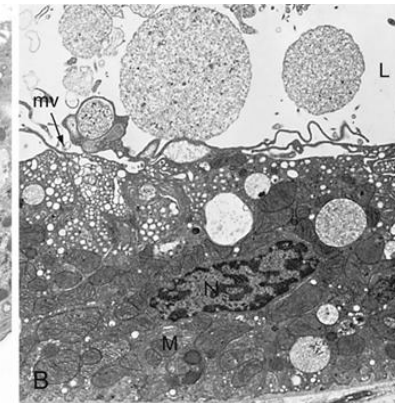
Normal kidney tubules



Reversible changes



Normal kidney tubules

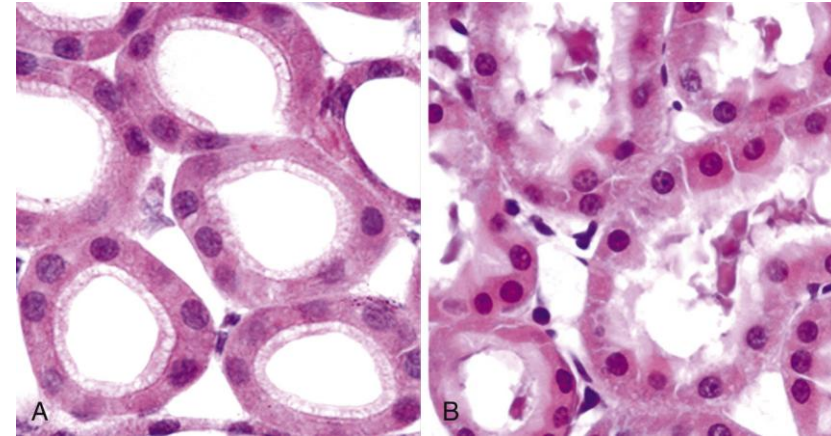


Reversible changes

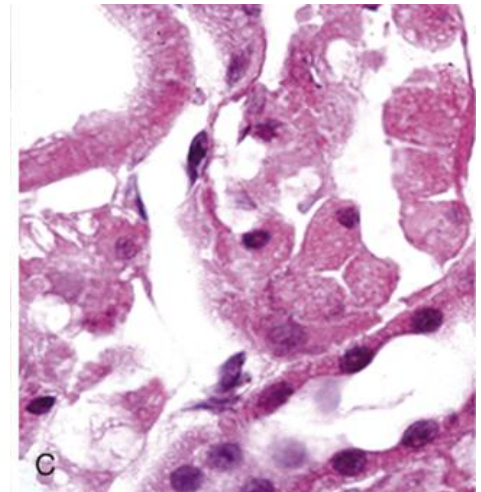
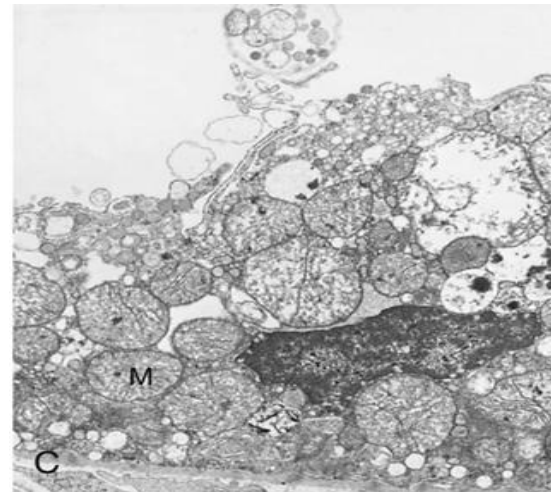
Progression of cell injury

Irreversible cell injury

- Plasma membrane-disrupted, leakage of contents
- Loss of nuclei
- Markedly swollen cells
- Markedly swollen mitochondria with electron-dense deposits
- Fragmentation of organelles

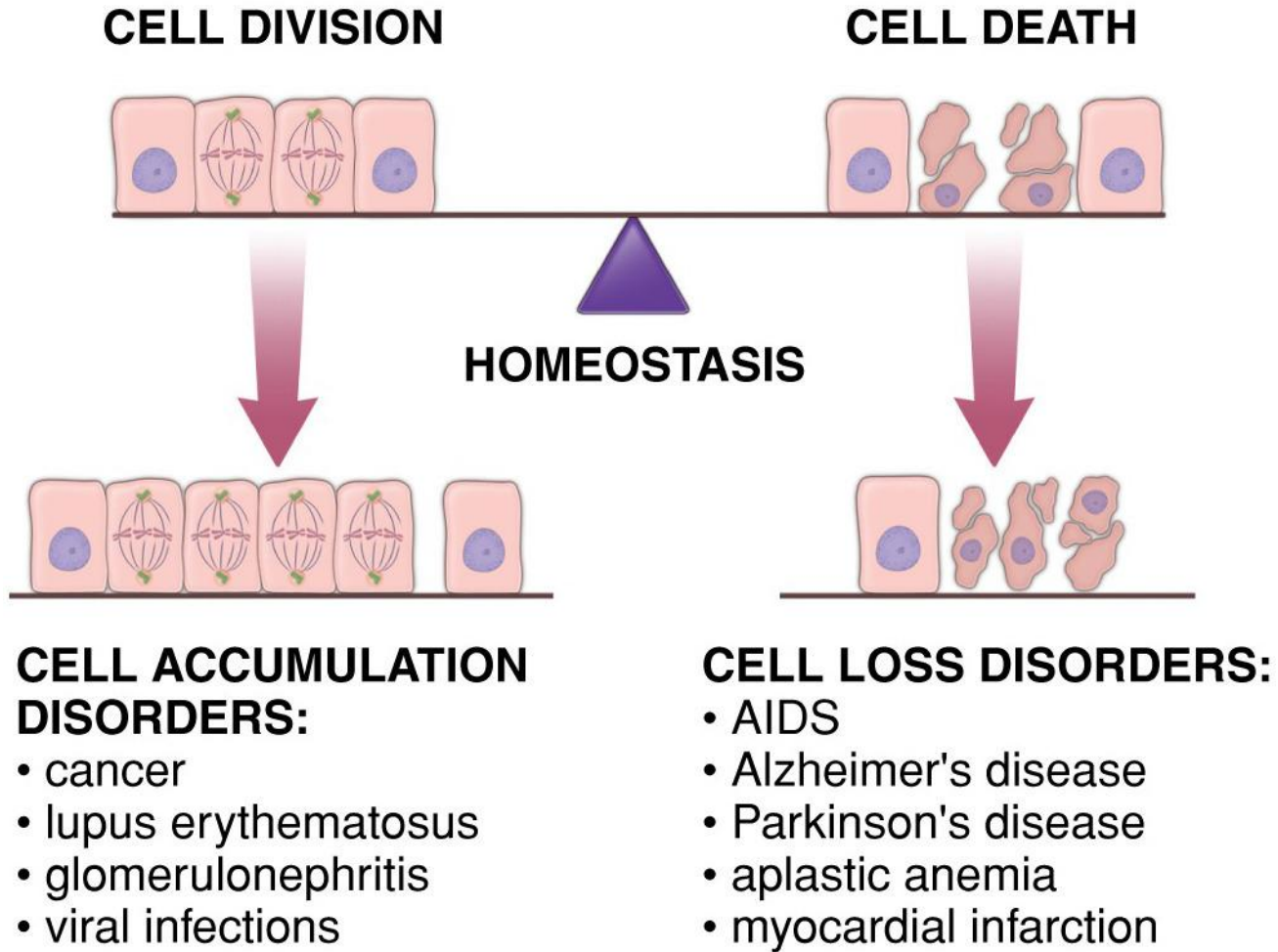


Normal kidney tubules



Irreversible injury

Apoptosis and Necrosis



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Cell Death

• 2 major mechanisms

1. Necrosis

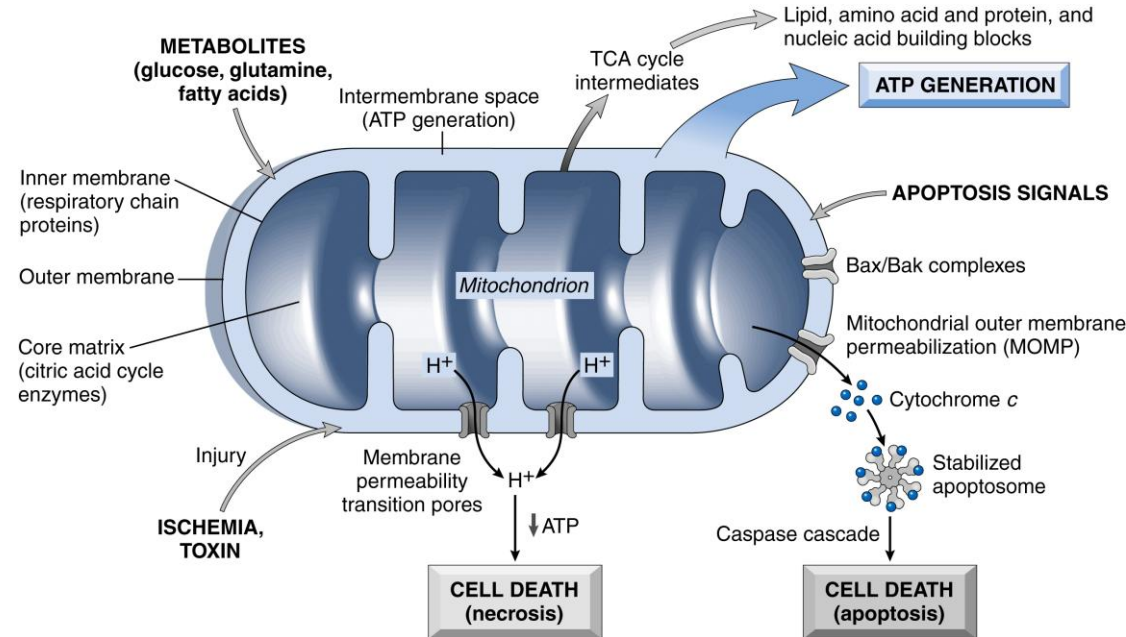
Cell death as a result of external injury.

a. Mitochondrial damage-development of mitochondrial permeability transition pores which dissipate proton gradient, prevent ATP generation

2. **Apoptosis**- programmed cell death triggered by:

- Extrinsic signals(cytotoxic T cells, inflammatory cytokines)
- Intrinsic pathways (DNA damage, Intracellular stress).
- Function- to remove unwanted cells (damaged, embryonic development)

- Distinct morphological differences
- Distinct biochemical differences



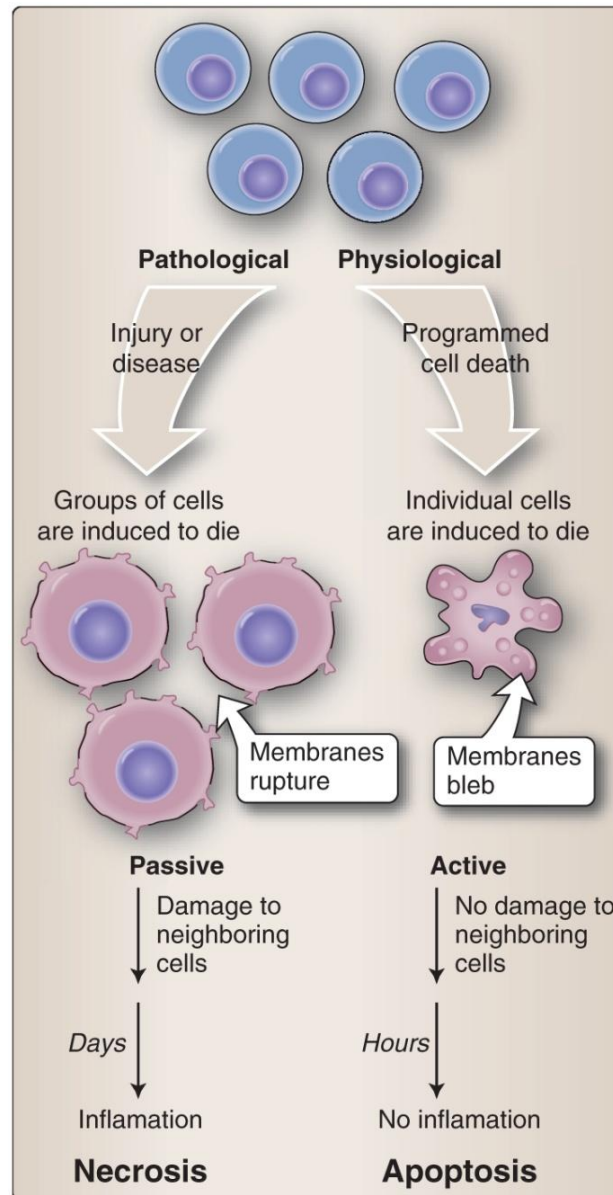
Cell Death

Necrosis

- Pathological
- Acute cell injury
- Cell unable to maintain homeostasis
- Cell swelling
- Loss of plasma membrane integrity
- Cell contents released
- Surrounding tissue damage
- Inflammation

Apoptosis

- Physiological
- Genetic
- Programmed cell death
- Cell shrinking
- DNA aggregation
- Maintains plasma membrane integrity
- No surrounding tissue damage
- No inflammation



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Cell Death

- 2 major mechanisms

- Necrosis**

- Principle outcome in many injuries
- Ex. Ischemia, toxins, infections & trauma

- Apoptosis**

- Regulated cell suicide program
- Occurs during development & throughout adulthood
- Physiological
 - Development
 - Tissue homeostasis
- Some pathological conditions
 - DNA damage
 - Misfolded proteins
 - Some viral infections

- Distinct morphological differences
- Distinct biochemical differences

Feature	Necrosis	Apoptosis
Cell Swelling	+++	-
Cell Shrinkage	-	+++
Damage to plasma membrane	+++	-
Plasma membrane blebbing	-	+++
Aggregation of chromatin	-	+++
Fragmentation of nucleus	-	+++
Oligonucleosomal DNA fragmentation	-	+++
Random DNA degradation	+	-
Caspase cascade activation	-	+++

Morphological differences

Biochemical features of Necrosis

- ATP depletion or decrease synthesis- secondary to hypoxia or ischemia
- Influx of Ca^{2+} and H_2O
- Activation of cytosolic enzymes (phospholipases- membrane degradation, proteases protein degradation)
- Inflammasome activation - ATP released from damaged cells
- Release of proinflammatory cytokines

Intracellular contents released can be measured in serum

Lactate dehydrogenase (LDH often used),

Troponin I (heart),

Creatine kinase (CK-MB),

Aspartate transaminase,

Alanine transaminase,

Alkaline phosphatase

Assessment of Necrosis

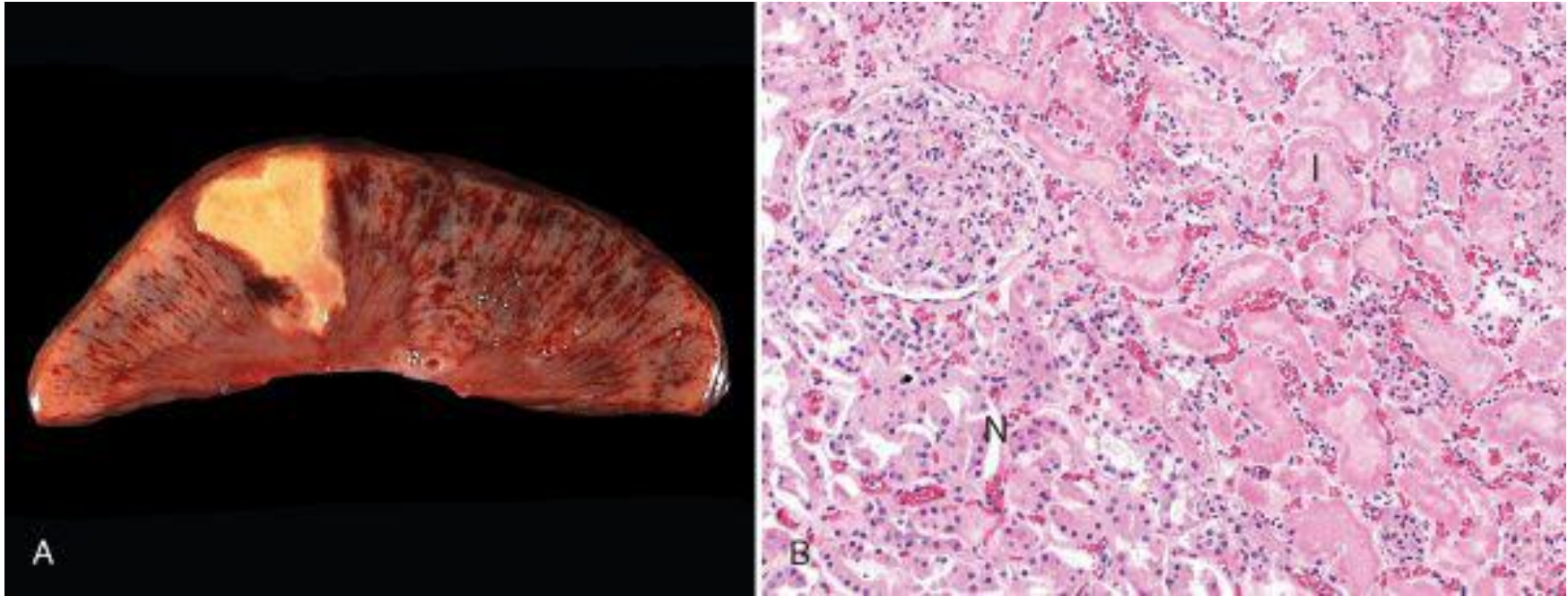
Morphological

- Propidium Iodide (PI) staining
 - Live cell imaging
 - PI intercalates & labels DNA
 - PI positive = leaky/discontinuous plasma membrane = necrosis
- H&E staining
 - Increased eosinophilia
 - Loss of cytoplasmic RNA
 - Increase in denatured proteins
 - Variable nuclear staining
 - Typical loss of basophilia & total loss of nuclear staining after a couple days; sometimes pyknosis visible
- TEM
 - Discontinuous plasma & organelle membranes

Biochemical

- Random DNA degradation
- Increased levels of lactate dehydrogenase (LDH)

Necrosis



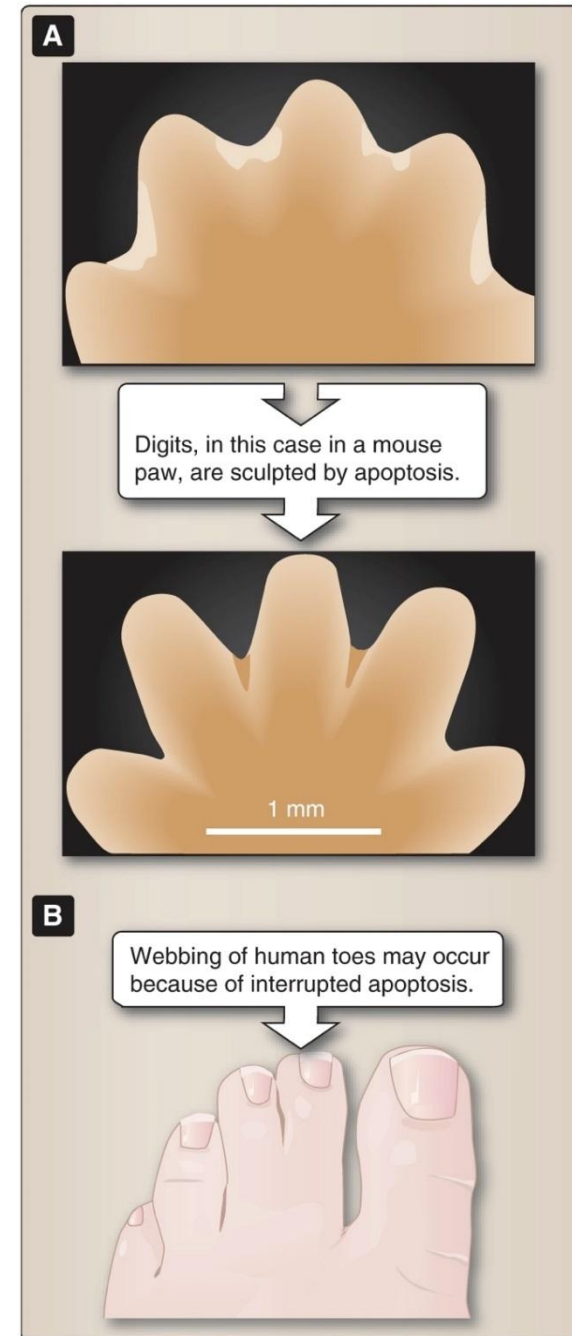
Coagulative necrosis of kidney

A. Gross

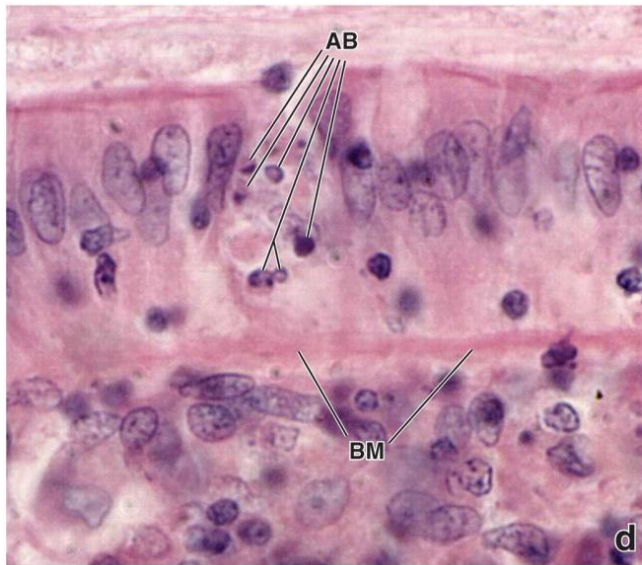
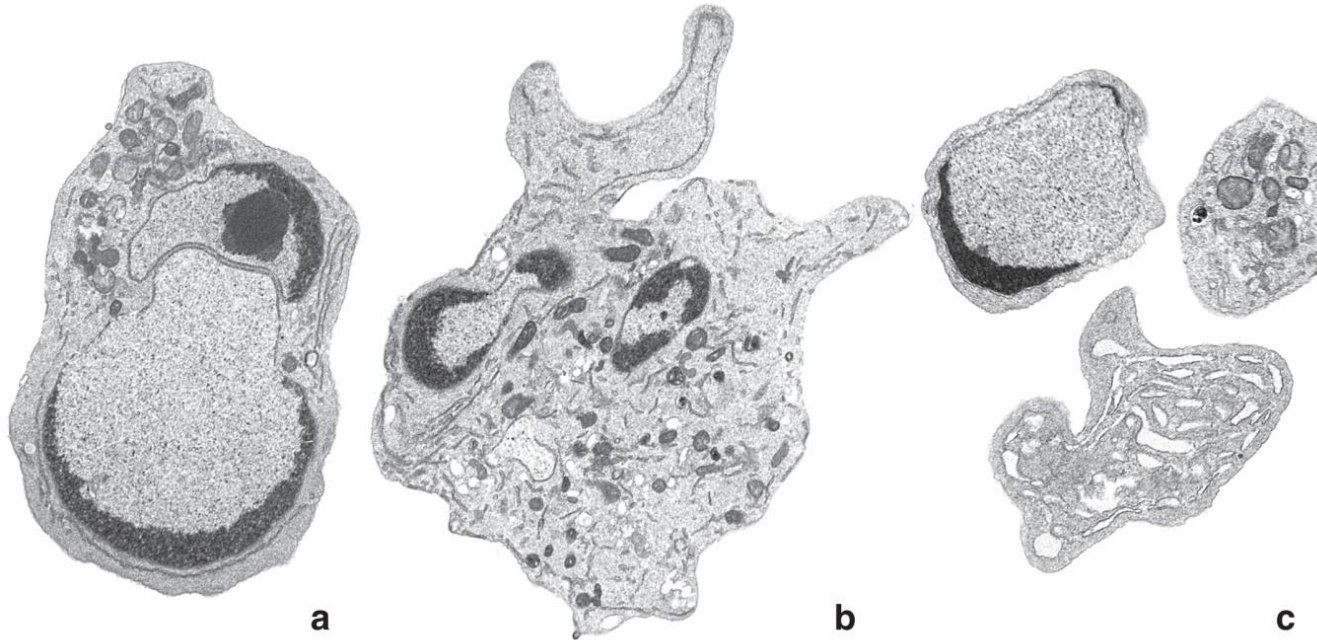
- Infarct (yellow)

Apoptosis

- Normal embryological development
 - Ex. Developing digits
- Normal tissue homeostasis
 - Billions of cells undergo apoptosis daily
 - Replaced by renewing cell populations
- Abnormal tissue homeostasis
 - Loss of cells due to apoptosis
 - Immune deficiency syndromes, some types of anemia
 - Irreplaceable due to non-renewing cell populations
 - Alzheimer's, Parkinson's, myocardial infarction
 - Accumulation of cells due to failure of apoptosis
 - Malignant neoplasia, autoimmune syndromes

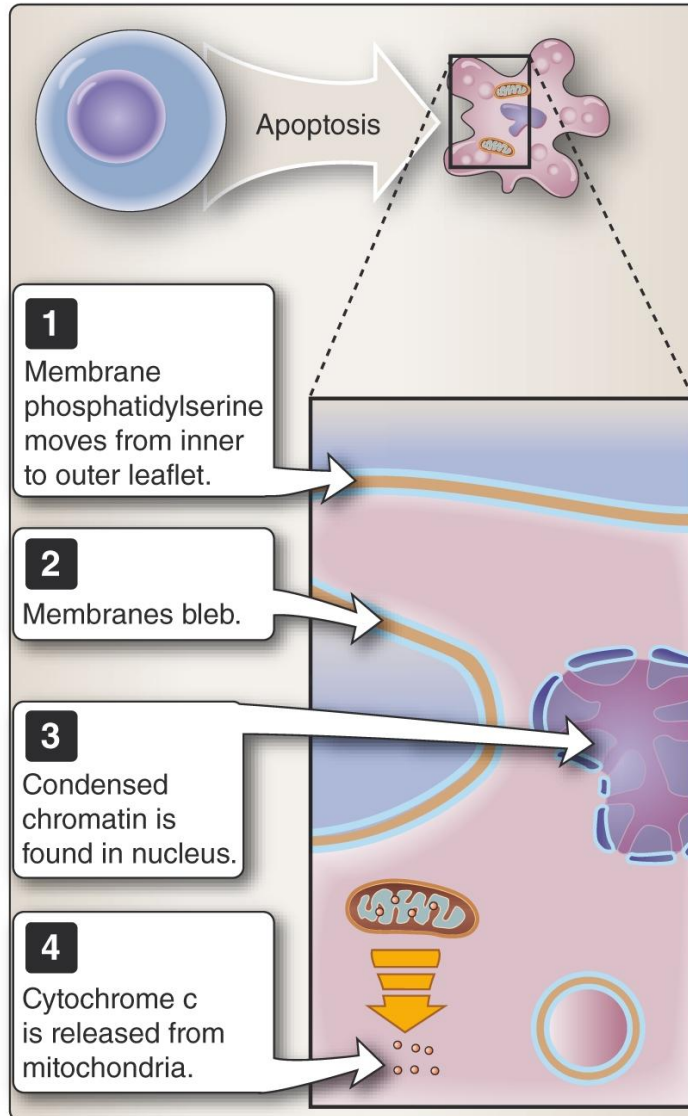


Apoptosis

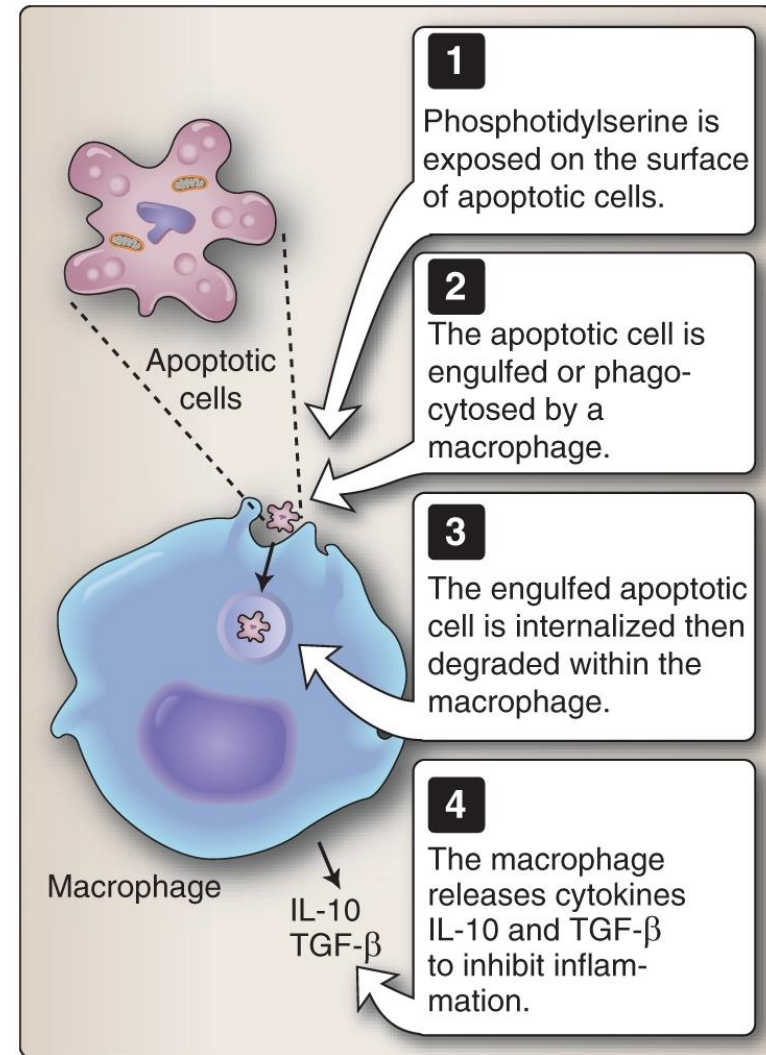


- a. Early apoptosis TEM x5,200
- b. Fragmentation of DNA TEM x5,200
- c. Apoptotic bodies TEM x5,200
- d. Intestinal epithelium H&E x750
 - Apoptotic bodies (AB)
 - Basement membrane (BM)

Apoptosis



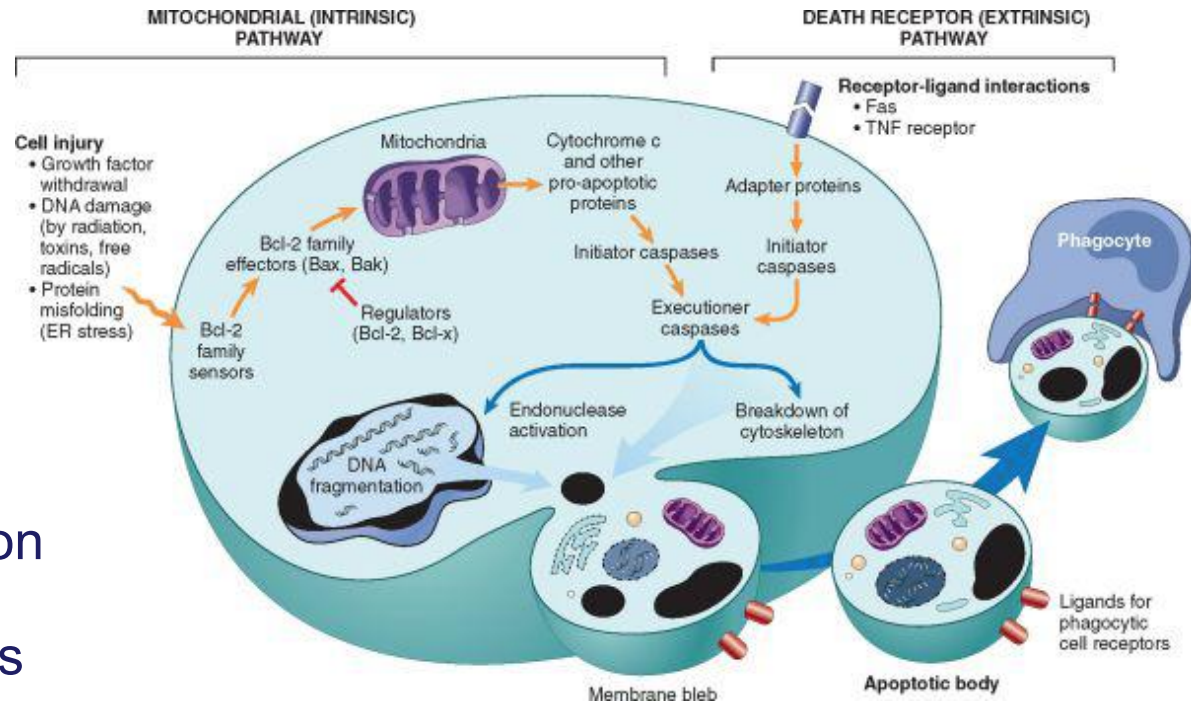
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Mechanisms of Apoptosis

- 2 pathways
 1. Intrinsic
 2. Extrinsic
- Differ in induction & regulation
- Both result in activation of initiator & executioner caspases

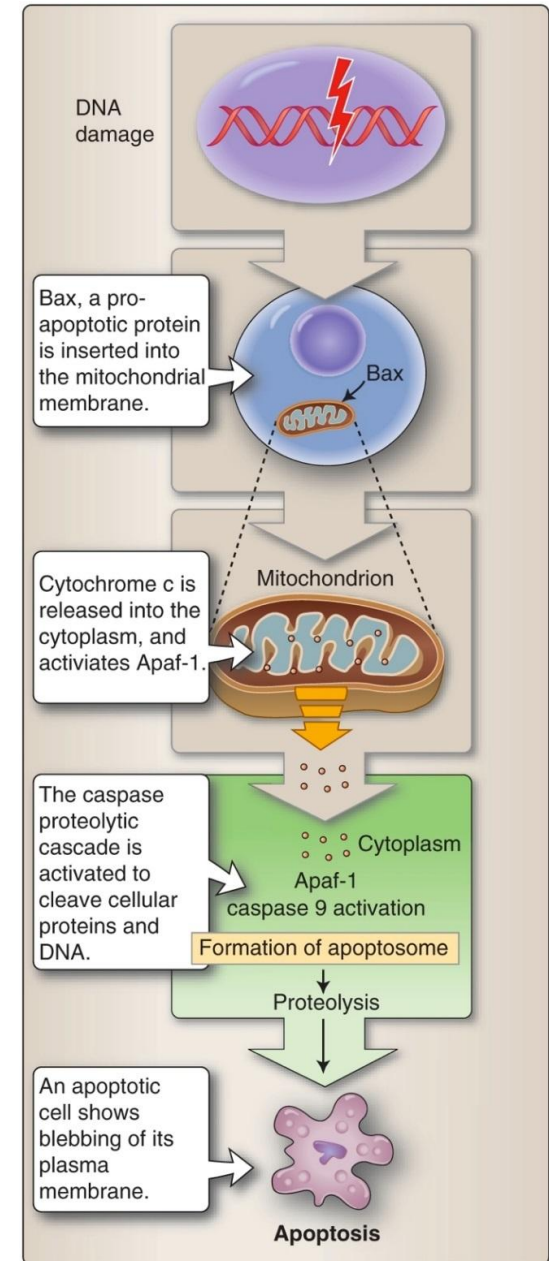


Robbins, 8th edition

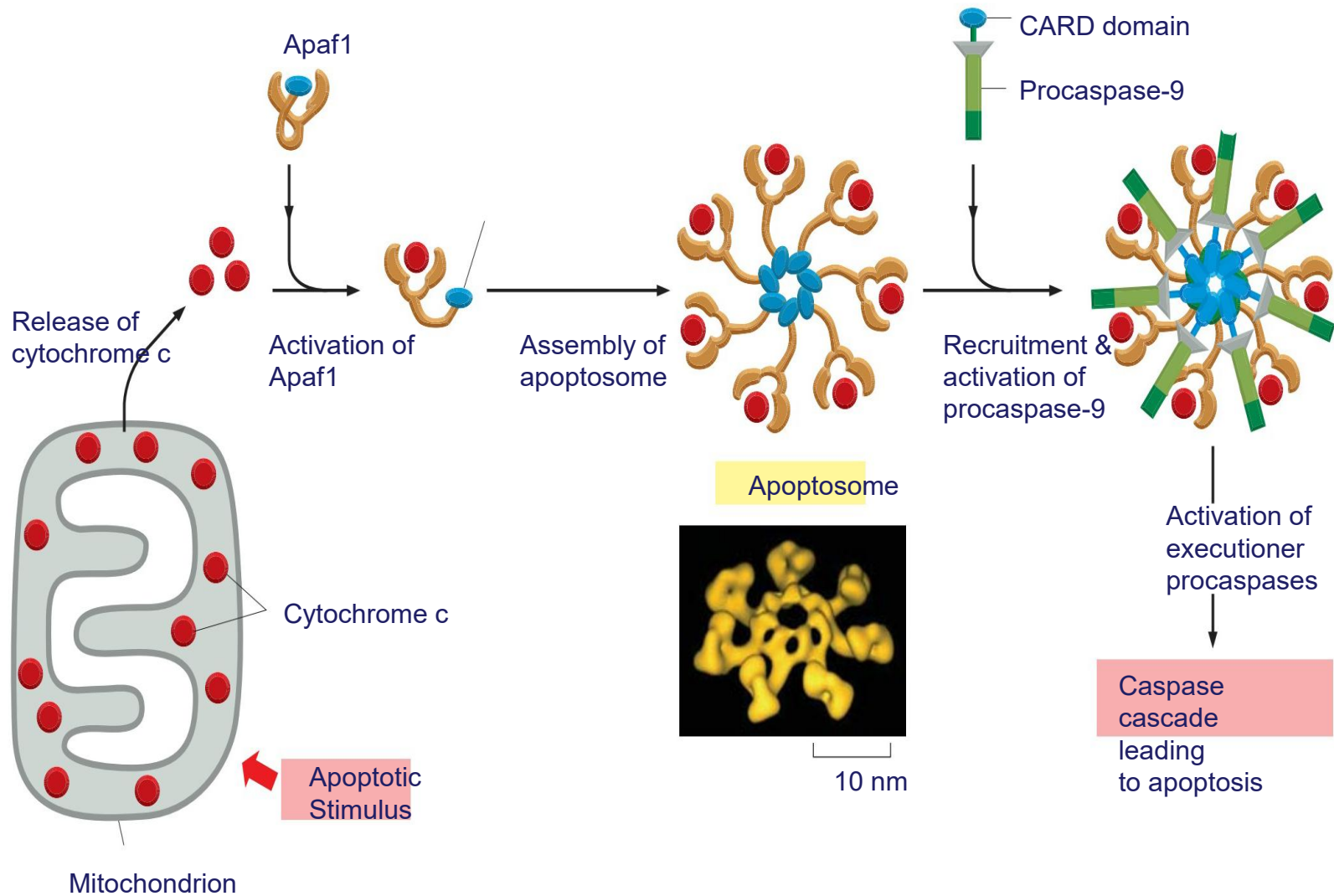
Apoptosis Pathways

1. Intrinsic

- Death signal
 - Ex. DNA damage
- Pro-apoptotic proteins upregulated
 - Ex. Bax
- Release of cytochrome c from mitochondria
- Apoptosome formation
 - Apaf-1
 - Apoptotic protease activating factor 1
 - Cytochrome c
 - Procaspase 9
- Caspase cascade
 - Activation of initiator caspase 9
 - Activation of effector caspases
 - Ex. Caspase 3



Intrinsic Pathway



Molecular Biology of the Cell, 5th edition

Bcl-2 Protein Family

- Intracellular regulators of apoptosis
- 2 functional classes
 - Pro-apoptotic
 - Bid
 - Links extrinsic & intrinsic pathways
 - Bax & Bak
 - Permeabilization of mitochondrial outer membrane
 - Anti-apoptotic
 - Bcl-2
- Balance of pro- & anti-apoptotic proteins largely determines

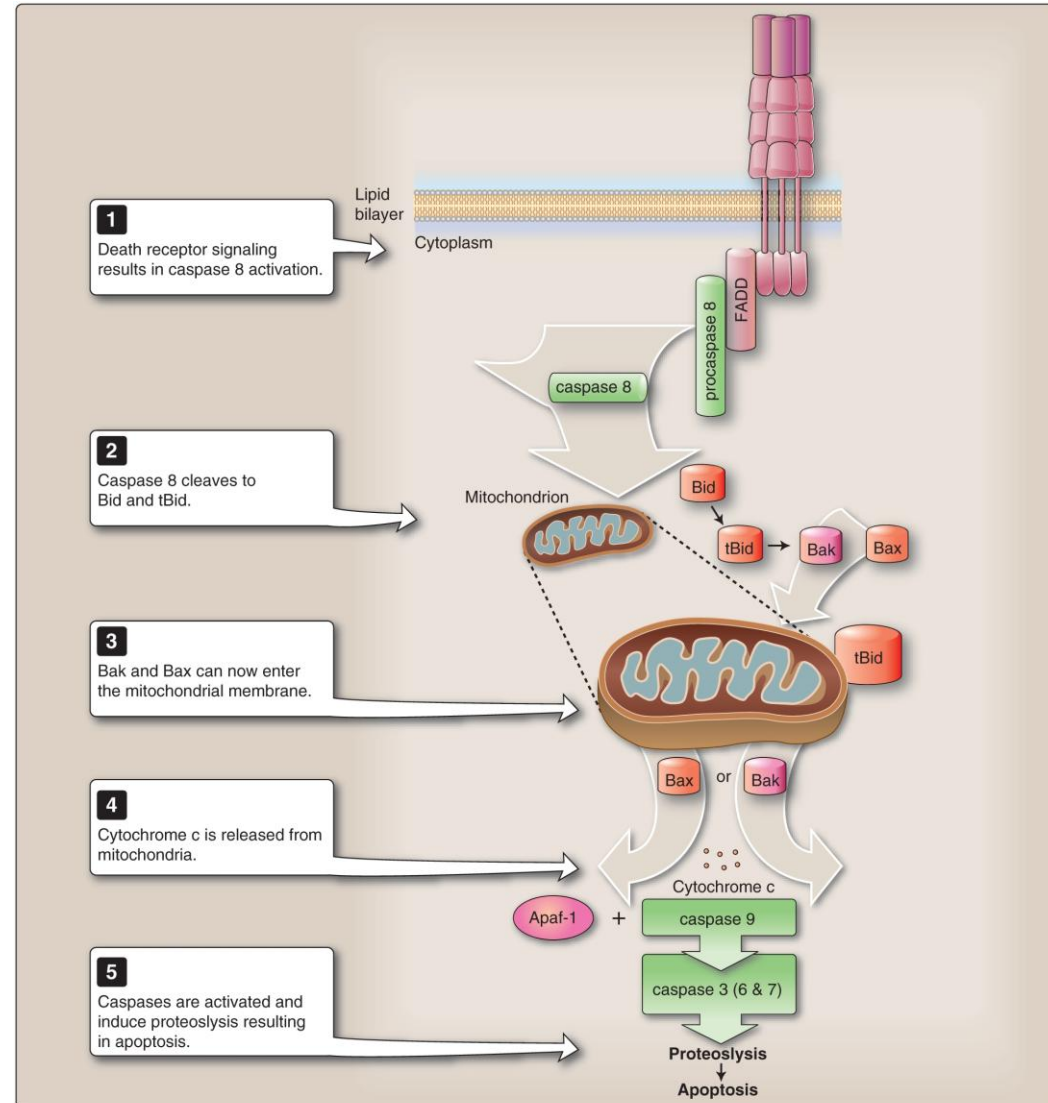
Bcl-2 Family

Pro survival:

Bcl-2
Bcl-xL

Pro death:

Bax
Bak
Bid

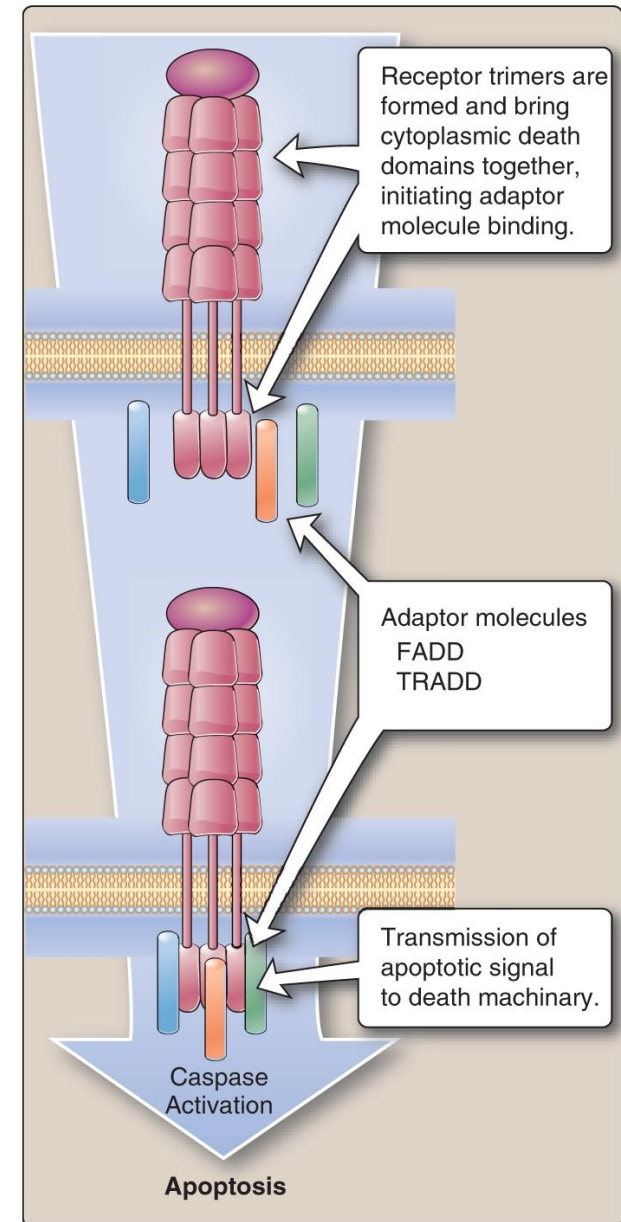


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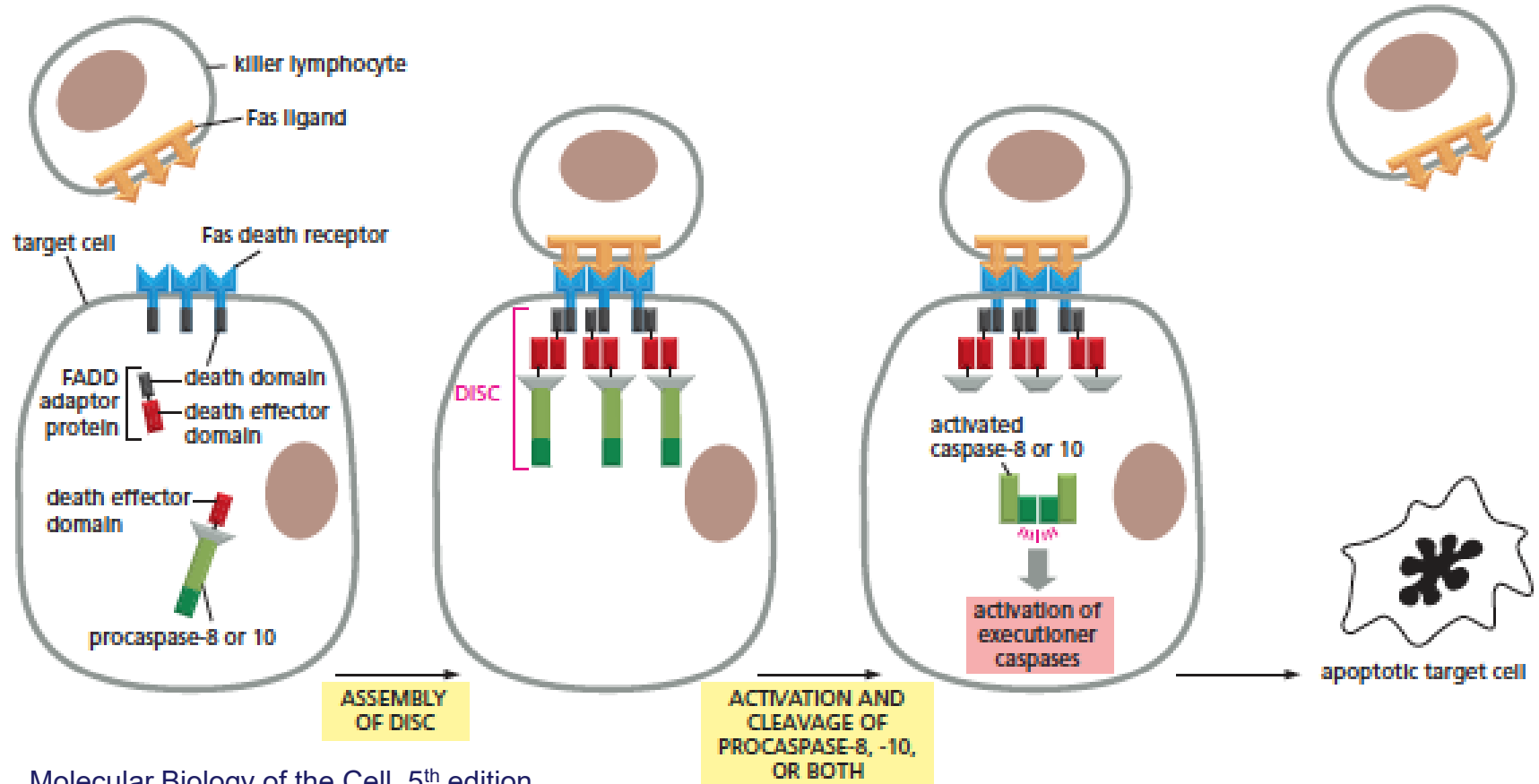
Apoptosis Pathway

2. Extrinsic

- Binding of ligand to death receptor
 - Fas ligand
- Recruitment of death domain adaptor proteins
 - FADD
 - TRADD
- Formation of death inducing signaling complex (DISC)
- Caspase cascade
 - Activation of initiator caspases
 - Ex. Caspase 8
 - Activation of effector or “executioner” caspases
 - Ex. Caspase 3



Extrinsic Pathway

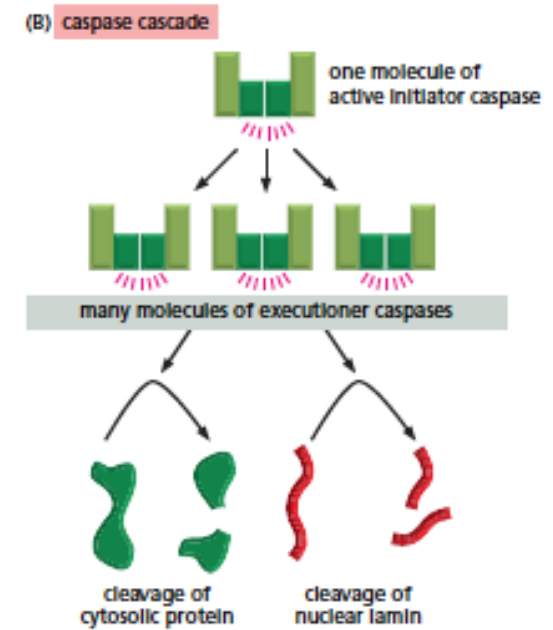
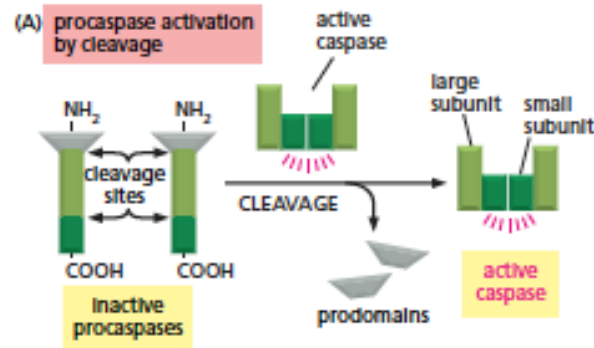


Molecular Biology of the Cell, 5th edition

Caspase Cascade

Caspases:	Role:
Caspase 1	Cytokine maturation
Apoptotic:	
Caspases 2,8,9,10	Initiator caspases
Caspases 3,6,7	Effector caspases
Caspases 4,5	Inflammation
Caspase 14	Skin development

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Molecular Biology of the Cell, 5th edition

- **Caspases**
 - Family of protease enzymes
 - Cysteine-dependent aspartate-directed proteases
 - Synthesized as inactive precursors – procaspases
 - Activated by proteolytic cleavage
 - Target cytoplasmic & nuclear proteins and caspase-active DNase

Biochemical features of Apoptosis

Apoptosis

- Protein cleavage- effected by caspases, proteins are cleaved at aspartic acid residues
- Protein cross-linking- through the expression and activation of transglutaminases
- DNA breakdown - Mg^{2+} and Ca^{2+} endonucleases break down DNA to fragments of 180 – 200 base pairs (DNA laddering)
- Phagocytic recognition-the expression of phosphatidylserine on the outer layer of the plasma membrane

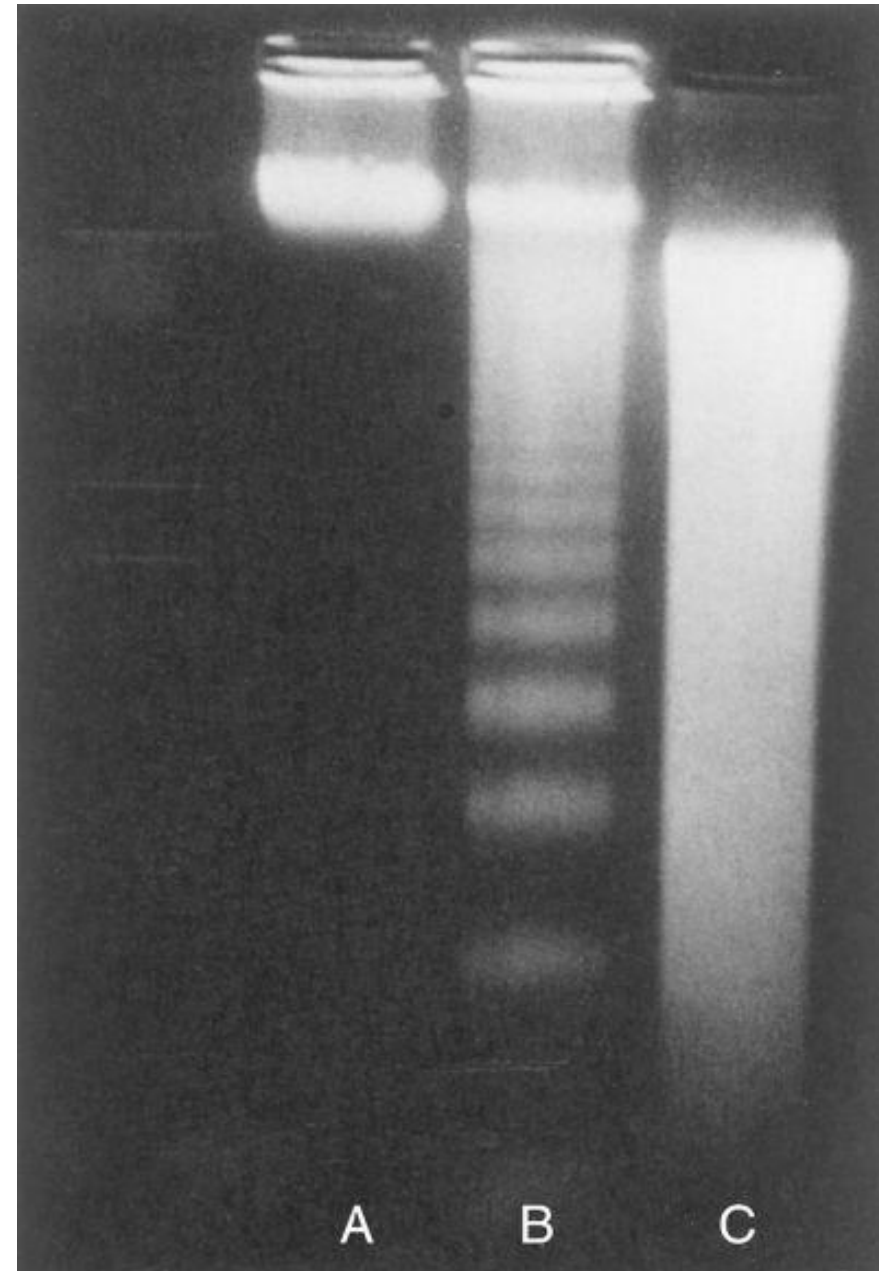
Assessment of Apoptosis

- Caspase activity
- Annexin 5
 - Protein binds to phosphatidylserine when exposed on the outer leaflet of plasma membrane
- DNA Laddering
 - DNase cleaves internucleosomal DNA
 - 180-200 bp fragments
- TUNEL
 - **t**erminal **u**ridine deoxynucleotidyl transferase nick **e**nd **l**abeling
 - Detects DNA fragmentation

DNA Laddering

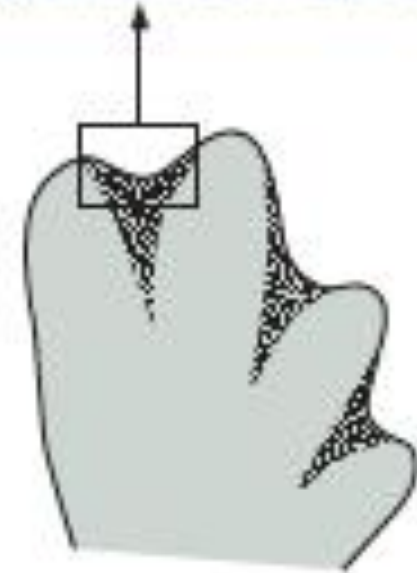
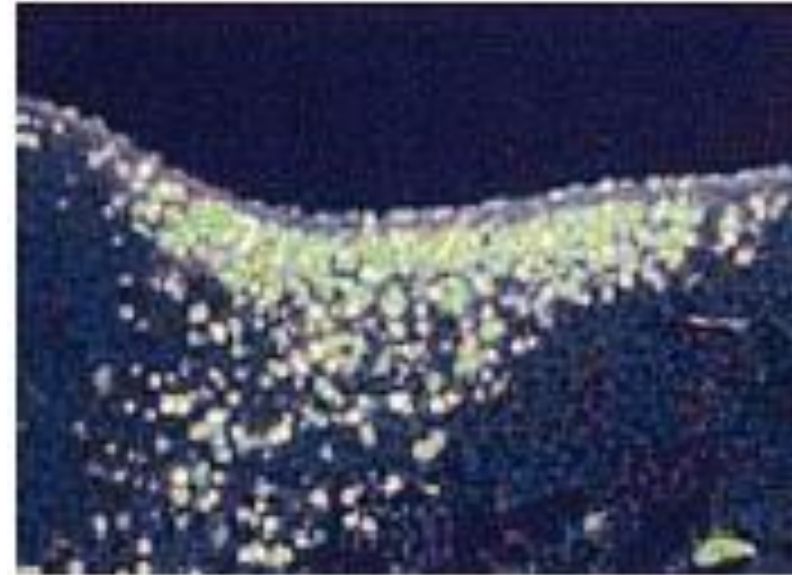
Agarose gel electrophoresis of DNA from cultured cells

- A. Viable cells
- B. Apoptotic cells
- C. Necrotic cells



Tunel

- Assay detects DNA fragmentation
- Tagged or labeled deoxynucleotides to 3' DNA ends



(B)